

DINGANI MUNSAKA

Berea, KY | 859-646-4885 | munsakad@berea.edu | [linkedin.com/in/dingani-munsaka](https://www.linkedin.com/in/dingani-munsaka) | github.com/munsakad | munsakad.github.io

SUMMARY

Software and hardware engineer who has shipped production Python, JavaScript, and Next.js systems adopted by 4,000+ students, supervisors, and staff across three internships; architected normalized MySQL schemas and Dockerized deployments; and built AI-grounded RAG, quiz, and BI platforms spanning 1,000+ curated records and 100+ learning modules.

EXPERIENCE

Web Application Engineer Intern

Jun 2026 – Aug 2026

Student Software Development Team, Berea College

Berea, KY

- Re-architected the Labor Status Form portal (Python, JavaScript, MySQL) into a unified per-department allocation workflow, replacing fragmented per-role tracking screens and driving adoption by 1,200+ students and 500+ supervisors across 200+ departments.
- Containerized the full production stack with Docker on Linux (Ubuntu/WSL), collapsing a manual multi-step install into a single `docker compose up` deployment used by the engineering team.
- Designed the entity-relationship model and normalized MySQL schema (users, departments, labor positions, allocations) that shipped directly to production, eliminating duplicate student records campus-wide.
- Diagnosed and resolved responsive layout defects on the CELTS community-service platform (500+ users), restoring successful form submissions on mobile and tablet devices.

Full Stack Product Engineer Intern

Jun 2026 – Aug 2026

Cardiora Software House

Remote

- Shipped three production Next.js/MySQL systems end-to-end: a multi-school student platform serving 2,500+ users, a role-based employee management and analytics dashboard, and an event platform with QR-code ticketing and integrated payments.
- Built a portfolio management platform (user profiles, project/resume uploads, admin moderation) and delivered every release through Git feature branches, peer code review, and two-week agile sprints.

AI & Full Stack Engineering Fellow

Jun 2025 – Aug 2025

DevWeekends

Remote

- Built two production Next.js/React applications: an AI study platform generating quizzes, flashcards, and personalized recommendations, and an AI resume builder with ATS scoring, cover-letter generation, and PDF export.
- Engineered an AI coding assistant for multi-language code generation, bug detection, and optimization, serving PyTorch models to a React/Next.js frontend through a REST API.

PROJECTS

Distributed Cloud File Storage

Sep 2024 – Jan 2025

TypeScript, gRPC, Kubernetes, Terraform, AWS S3

[GitHub](#)

- Engineered a gRPC-based distributed storage service that chunks files into 1 MiB pieces, encrypts each with AES-256-GCM, and replicates them across multiple local and S3 targets with automatic failover reads.
- Implemented a rollback-able versioning system so re-uploads create new versions instead of overwriting data, and automated deployment with Kubernetes manifests and Terraform-provisioned AWS infrastructure.

Cortex: AI Business Intelligence Platform

Feb 2026 – May 2026

Python, FastAPI, SQLite, Gemini API, React

[GitHub](#)

- Built an AI-assisted BI platform that profiles an uploaded CSV and uses Gemini to generate dashboards, SQL queries, reports, and forecasts grounded in real computed results rather than hallucinated figures.
- Implemented a validated, read-only SQL execution layer so every Gemini-generated query runs as a real statement against SQLite, and added a NumPy-based forecasting module with AI-written trend commentary.

Hardware Engineering Education Hub

May 2025 – Apr 2026

Python, MySQL, JavaScript, HTML/CSS

[GitHub](#)

- Built a full-stack quiz platform with 100+ learning modules, auto-graded quizzes, and a CMS that lets instructors publish new material without touching code.
- Designed a normalized MySQL schema and dynamic frontend supporting multi-topic filtering and per-user progress tracking, so returning students resume at their last completed module.

DMHardware AI Chatbot

Aug 2025 – Jan 2026

Python, Gemini API, PyTorch, Scikit-Learn, Pandas

[GitHub](#)

- Grounded a Gemini-API chatbot on 1,000+ custom-curated records covering electronics, robotics, PCB design, and digital logic so responses cite course-specific material.
- Built the preprocessing and retrieval pipeline (Pandas, NumPy, Scikit-Learn) that cleans the record set and selects relevant context per query, keeping answers anchored to curated domain data.

EDUCATION

Berea College

Aug 2024 – Expected May 2028

B.A. Computer Science & B.S. Engineering Technologies

Berea, KY

- GPA: 3.90/4.00 | Dean's List (4x) | Full-Ride No Tuition Promise Scholarship (\$263,000)
- Relevant Coursework: Data Structures & Algorithms, Distributed Systems, Operating Systems, Database Systems, Machine Learning